RK300-08 Communication Protocol（MODBUS-RTU）

1.Communication Parameters (factory default):

Baud rate: 9600bps, Data bits:8, Stop bit:1, Check bit:no, Address:0x01

2.Examples for read data(PM2.5)

Host Scan Order：

Slave id	Function code	Address_H	Address_L	Quantity_H	Quantity_L	CRC_L	CRC_H
0x01	0x03	0x00	0x00	0x00	0x01	0x84	0x0A

Sensor Response:

Slave id	Function code	Number of bytes	PM2.5		CRC_L	CRC_H
0x01	0x03	0x02	0x01	0x43	0xF8	0x25

PM2.5: (0143)H=(323)D,332μg/m^3

3.Examples for modify the address

Host Scan Order(Changed to 02H)：

Slave id	Function code	Address_H	Address_L	New id_H	New id_L	CRC_L	CRC_H
0x00	0x06	0x00	0x0F	0x00	0x02	0x39	0xD9

Sensor Response:

Slave id	Function code	Address_H	Address_L	New id_H	New id_L	CRC_L	CRC_H
0x02	0x06	0x00	0x0F	0x00	0x02	0x38	0x3B

4.Examples for modify the baud rate(The changes require a restart to take effect)

Host Scan Order(Changed to 19200)(Restart after modifying the baud rate to 19200)：

Slave id	Function code	Address_H	Address_L	New id_H	New id_L	CRC_L	CRC_H
0x00	0x06	0x00	0x28	0x00	0x02	0x89	0xD2

Sensor Response:

Slave id	Function code	Address_H	Address_L	New id_H	New id_L	CRC_L	CRC_H
0x01	0x06	0x00	0x28	0x00	0x02	0x88	0x03

5.Examples for modify the Parity(The changes require a restart to take effect)

Host Scan Order(Changed to EVEN)(Even parity enabled after modification & restart)：

Slave id	Function code	Address_H	Address_L	New id_H	New id_L	CRC_L	CRC_H
0x00	0x06	0x00	0x2A	0x00	0x02	0x28	0x12

Sensor Response:

Slave id	Function code	Address_H	Address_L	New id_H	New id_L	CRC_L	CRC_H
0x01	0x06	0x00	0x2A	0x00	0x02	0x29	0xC3

6.Register Description

Register Name	Function Code	Register Address	Data Type	Byte Length	R/W	Resolution（Standard）	Note
PM2.5	0x03	0x0000	Int16	2	R	1μg/m^3	/
PM10	0x03	0x0001	Int16	2	R	1μg/m^3	/
TSP	0x03	0x0002	Int16	2	R	1μg/m^3	/
Air Temperature	0x03	0x0003	Int16	2	R	0.1℃	/
Air Humidity	0x03	0x0004	Int16	2	R	0.1%RH	/
Air Pressure	0x03	0x0005	Int16	2	R	0.1hPa	/
CO_2	0x03	0x0006	Int16	2	R	1ppm	/
O_2	0x03	0x0007	Int16	2	R	0.1%VOL	/
SO_2	0x03	0x0008	Int16	2	R	0.01ppm	/
NO2	0x03	0x0009	Int16	2	R	0.01ppm	/
O_3	0x03	0x000A	Int16	2	R	0.01ppm	/
CO	0x03	0x000B	Int16	2	R	1ppm	/
H_2S	0x03	0x000C	Int16	2	R	0.01ppm	/
NH_3	0x03	0x000D	Int16	2	R	0.01ppm	/
TVOC	0x03	0x000E	Int16	2	R	0.001ppm	/
Sensor Address	0x06	0x000F	Int16	1	W	/	/
CH_4	0x03	0x0010	Int16	2	R	0.1%LEL	/
CL_2	0x03	0x0011	Int16	2	R	0.01ppm	/
PM1.0	0x03	0x0012	Int16	2	R	1μg/m^3	/
CH_2O	0x03	0x0013	Int16	2	R	0.01ppm	/
Odour	0x03	0x0014	Int16	2	R	0.01ppm	/
NO	0x03	0x0015	Int16	2	R	0.01ppm	/
Illuminance	0x03	0x0016	Int16	2	R	10Lux	/
Civil Gas	0x03	0x0017	Int16	2	R	0.1%LEL	/
Air Quality	0x03	0x0018	Int16	2	R	0.01mg/m^3	/
Dew Point Temperature	0x03	0x0019	Int16	2	R	0.1℃	/
HC	0x03	0x001A	Int16	2	R	0.01ppm	/
Noise	0x03	0x001B	Int16	2	R	0.1db	/
N_2O	0x03	0x001C	Int16	2	R	0.01%VOL	/
Baud Rate Selection	0x06	0x0028	Int16	2	W	/	1: 9600 2: 19200 3: 38400 4: 57600 5: 115200
Baud Rate	0x03	0x0029	Int16	2	R	/	Default:9600
Parity Selection	0x06	0x002A	Int16	2	W	/	0: NONE 1: ODD 2: EVEN